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A Candidate Third Transiting Planet on an 18.9-day Orbit around the M Dwarf TOI-4342, Matching a Dynamical Prediction

Ricardo Pinho[1]

[1] *Independent researcher*

ABSTRACT

I report a candidate third transiting planet around TOI-4342 (TIC 354944123), an M0 dwarf hosting two sub-Neptunes at 5.54 and 10.69 days. Nine sectors of TESS 2-minute photometry show a 1182 ± 107 ppm, 2.4 hr transit every 18.908226 days, $R_p = 2.25 \pm 0.13\,R_\oplus$, in all ten observed windows (signal-to-noise ratio 11.7). Period, radius and inclination match the planet predicted for this system by E. V. Turtelboom et al. (2025). Difference images place the source within $11''$ of the target and exclude every Gaia neighbor able to mimic it. With a speckle contrast curve and the pixel-excluded neighbors removed, TRICERATOPS gives a false-positive probability of 0.018, just above the 0.015 validation threshold, and a nearby-false-positive probability of zero. The three planets are coplanar to within 1° and dynamically stable. I provide an ephemeris for ground-based confirmation.

Keywords: Exoplanet detection methods (489) — Exoplanets (498) — M dwarf stars (982) — Transit photometry (1709)

1. INTRODUCTION

TOI-4342 ($T = 11.03$, 61.5 pc; $T_{\rm eff} = 3866 \pm 92$ K, $R_\star = 0.598 \pm 0.018\,R_\odot$, $M_\star = 0.587 \pm 0.013\,M_\odot$) hosts planets b ($P = 5.538$ d) and c (10.689 d) near the 2:1 resonance (E. Tey et al. 2023), with masses 7.3 ± 1.3 and $4.8 \pm 1.4\,M_\oplus$ (L. Parc et al. 2026). E. V. Turtelboom et al. (2025) predicted an additional planet at $18.5^{+2.7}_{-2.0}$ d with $2.24\,R_\oplus$. TOI-4342 entered my search for additional transits around small-star planet hosts.

2. DATA AND DETECTION

TESS (G. R. Ricker et al. 2015) observed TOI-4342 in Sectors 13, 27, 66, 67, 93, 94 and 101–103 (2019–2026). I used SPOC 2-minute PDCSAP light curves and target pixel files (J. M. Jenkins et al. 2016), 20-second light curves (Sectors 93–103), and QLP (C. X. Huang et al. 2020) and TESS-SPOC (D. A. Caldwell et al. 2020) full-frame-image light curves. Each sector was detrended with a 0.75-day biweight filter (M. Hippke et al. 2019) with all transits masked, and models of b and c were subtracted. A box least-squares search (G. Kovács et al. 2002) found the 18.9-day signal.

Fourteen predicted transits fall within the sectors; ten have data, including one in cadences that SPOC excludes from its planet search and one partial transit. All ten show the transit (Figure 1b): depths agree ($\chi^2 = 8.7$ for 9 degrees of freedom), as do the 2019–2023 and 2025–2026 halves (0.98 ± 0.11 and 0.89 ± 0.12 of the model depth). The signal is present in simple-aperture flux (10.7σ) and in QLP (8.8σ). Applied to the same data with the real windows removed, the identical statistic never reached the observed value at 3,000 random periods (maximum 8.9 against 13.4). Depths at $P/2$, $P/3$ and $P/4$ are consistent with zero (53 ± 78 ppm at $P/2$), and odd and even transits agree (0.5σ), which excludes $2P$. Transit times are consistent with a linear ephemeris ($\chi^2 = 7.9$, 6 degrees of freedom).

A `batman` (L. Kreidberg 2015) and `emcee` (D. Foreman-Mackey et al. 2013) fit with a stellar-density prior of 3.86 ± 0.36 g cm^{-3} gives $P = 18.908226 \pm 0.000042$ days, $T_0 = 2459904.5666 \pm 0.0021$ BJD$_{\rm TDB}$, $R_p/R_\star = 0.0344 \pm 0.0016$, $b = 0.76 \pm 0.03$, $i = 88.96 \pm 0.07°$, $a/R_\star = 41.7 \pm 1.3$ and a duration of 2.44 ± 0.11 hr (Figure 1a). Fitted freely, the implied density, $8.8^{+4.7}_{-5.9}$ g cm^{-3}, is consistent with the star. The inclination matches b ($88.83 \pm 0.22°$) and c ($89.61 \pm 0.26°$).

3. VETTING

Gaia DR3 (Gaia Collaboration et al. 2016, 2023) shows no sign of a companion (RUWE 0.98). Gemini-South/Zorro speckle imaging (E. Tey et al. 2023; instrument: N. J. Scott et al. 2021) detected none at $\Delta m_{832} > 4.7$ beyond $0.1''$.

On a 1976 photographic plate, when the star was 7.4″ from its 2026 position, no source appears at that position to $\Delta T \approx 7$; a 1182 ppm signal requires $\Delta T < 7.4$.

For each sector I fitted the in-transit difference image with that sector's out-of-transit image, shifted across a sky grid, and summed χ^2 over sectors. On the same pixels, b and c give target $\Delta\chi^2$ of 9.5 and 3.5. The candidate gives 11.4, a 3σ region within 11″ of the target, and $\Delta\chi^2 \geq 79$ for every Gaia star that could mimic it (Figure 1c). Its depth versus aperture size tracks those of b and c.

4. STATISTICAL VALIDATION AND DYNAMICS

TRICERATOPS (S. Giacalone et al. 2021), with SPOC apertures from nine sectors and the Zorro contrast curve, gives a false-positive probability (FPP) of 0.0161 ± 0.0002 and a nearby-false-positive probability (NFPP) of 0.0005. Removing the pixel-excluded neighbors gives FPP $= 0.018 \pm 0.002$ and NFPP $= 0$. Adding MOLUSC (M. L. Wood et al. 2021) limits on bound companions from the imaging and Gaia gives 0.016 ± 0.011. The remainder is almost entirely a planet transiting an unresolved bound companion. The candidate therefore sits at, not below, the validation threshold of FPP < 0.015 and NFPP < 0.001 (S. Giacalone et al. 2021), and I do not claim validation. This calculation does not include the lower false-positive rate of candidates in systems that already host transiting planets, the "multiplicity boost" used to validate Kepler multis (J. J. Lissauer et al. 2012, 2014); that factor would lower the FPP further, but it was calibrated on Kepler's population, and I do not apply it here.

N-body models predict transit-timing variations of a few minutes, below the TESS timing precision of ≈8 minutes per transit; the observed times neither require nor exclude them. SPOCK (D. Tamayo et al. 2020) with REBOUND (H. Rein & S.-F. Liu 2012) gives a stability probability of 0.89 (median over mass and eccentricity draws; 97% of draws above 0.5) for b, c and the candidate, including the 47.5-day radial-velocity candidate of L. Parc et al. (2026).

5. A SECOND, WEAKER SIGNAL

The same data show a 287 ± 40 ppm transit every 1.957690 days ($1.11\,R_\oplus$, signal-to-noise ratio 8.8), consistent across seasons, with no odd–even difference and with the expected density. Its pixel localization is looser and TRICERATOPS gives FPP $= 0.013$ (0.007 with MOLUSC) and NFPP $= 0.001$, but its signal-to-noise ratio is close to the level at which my search expects noise detections. I regard it as a lower-confidence candidate.

6. FOLLOW-UP

TESS will not observe TOI-4342 again through Sector 134. A 1.2 ppt, 2.4 hr transit is within reach of 1-m-class telescopes. Predicted mid-transit times are $T_n = 2459904.5666 + 18.908226\,n$ $\mathrm{BJD_{TDB}}$, with uncertainty ≈12 minutes through 2027. Transits fully observable from the south include 2026 October 28 02:10 (Chile), 2027 May 24 01:56 (South Africa) and 2027 August 26 14:55 UTC (Australia).

7. DATA AND CODE

The TESS data are available at MAST: 2-minute light curves doi:10.17909/t9-nmc8-f686 and TESS-SPOC FFI light curves doi:10.17909/t9-wpz1-8s54. Code to reproduce this analysis is available at https://github.com/rpinho/planet-hunt (branch `round6-toi4342`).

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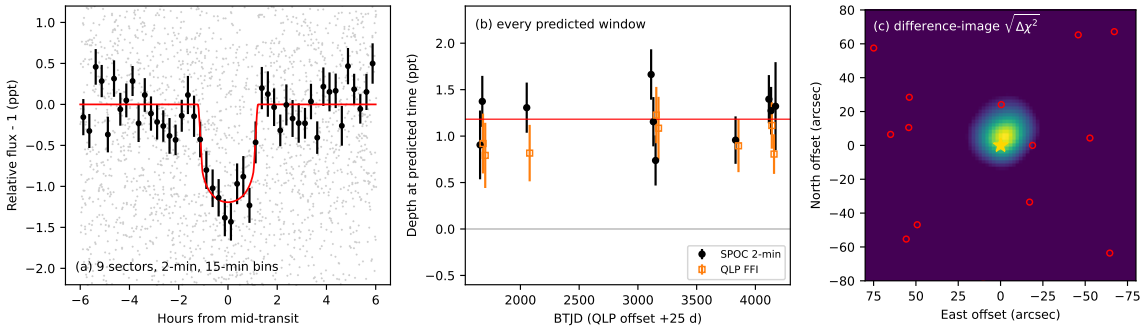


Figure 1. (a) SPOC 2-minute photometry of nine sectors, with b, c and the 1.96-day signal removed, folded on the 18.9-day ephemeris (15-minute bins), with the best-fit model. (b) Depth at each predicted transit in 2-minute (filled) and QLP (open, offset by 25 days) photometry; the line is the model depth. (c) Difference-image localization summed over nine sectors, $\sqrt{\Delta\chi^2}$ relative to the best position; star: TOI-4342; circles: Gaia stars bright enough to mimic the signal.

AI DISCLOSURE STATEMENT

Generative AI was used extensively. Claude Opus 5.5 (Anthropic) wrote and ran the analysis code, carried out the data retrieval, vetting, statistical and dynamical tests, searched the literature, and drafted the text, figure and bibliography.

Facilities: TESS, Gaia, Gemini:South (Zorro), ExoFOP

Software: astropy (Astropy Collaboration et al. 2022), batman (L. Kreidberg 2015), emcee (D. Foreman-Mackey et al. 2013), lightkurve (Lightkurve Collaboration et al. 2018), matplotlib (J. D. Hunter 2007), numpy (C. R. Harris et al. 2020), REBOUND (H. Rein & S.-F. Liu 2012), scipy (P. Virtanen et al. 2020), SPOCK (D. Tamayo et al. 2020), MOLUSC (M. L. Wood et al. 2021), tess-point (C. J. Burke et al. 2020), TRICERATOPS (S. Giacalone et al. 2021), wotan (M. Hippke et al. 2019)

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